

A Brief Survey on Ransomware with the Perspective of Internet Security Threat Reports

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Abstract— The cyber security has become a mandatory aspect in every industry. The cyber security companies publish threat reports on regular basis. However, their reports are based on internal data and cannot provide complete network attack trends. In this study, we present a survey on malware and ransomware attack trends by using the data obtained from multiple threat reports. We collect, process, and analyze the data to present it in the form of information. The presented information is useful for corporate executives and cyber insurance companies to examine the impact of malware and ransomware attacks in the recent years. As future prospect, we would like to use the presented information to formulate a cyber risk model of ransomware attack. We would like to develop a model to estimate the potential losses of an organization due to a successful ransomware attack.

Keywords— Ransomware Survey, Ransomware Attack, Malware Survey, Threat Reports, Survey on Threat Reports

I. INTRODUCTION

The ransomware attacks are exponential in the recent years and have huge impact on individuals and businesses all around the globe. The antivirus companies publish their quarterly and/or annually Internet Security Threat Reports (ISTR). However, each company publish reports based on their internal data, gathered from a specific antivirus agent to the clients worldwide. The purpose of our study is to provide a brief survey on ransomware attacks reported in ISTR for the years 2016-2020. This study provides a useful insight to corporate executives and security professionals in order to mitigate against the potential existing and zero-day ransomware attacks.

A malware is any malicious software that is used to cause damage to digital devices. Some of the malware types are virus, ransomware, and Trojan horse etc. An anti-virus is a software used to detect, prevent, and remove malware from digital devices. McAfee, Kaspersky, Symantec, Norton, and Malwarebytes are a few examples of anti-virus companies. The anti-virus companies publish their ISTR on certain intervals. Their purpose of publishing the threat reports is to let the people know about recent network attack trends. These trends are useful for general public to keep them safe from cyber-attacks, and for security professionals to motivate them to design solutions to mitigate cyber-attacks. In the presented work, we provide a survey on ransomware attacks using the data obtained from ISTR published by McAfee, Kaspersky, and Symantec.

The rest of the paper is organized as follows: the related work is presented in section II. The research methodology and the scope of the work are presented in section III. The section IV provide the malware attack trends. The ransomware attack trends are presented in section V. Finally, the conclusion and future directions are discussed in section VI.

II. RELATED WORK

The purpose of the related work is to find out research done on ransomware attacks. Being an important topic, a lot of research has been taken place on causes of ransomware attacks, evolution of ransomware, and detection and prevention techniques of ransomware attacks.

A. Causes of Ransomware Attacks

According to Statista [31], there are five major attack vectors that become the main causes of ransomware attacks. The spam or phishing emails are 67% successful in injecting ransomware to the system. The lack of cyber security training become the reason of 36% successful attacks. The weak passwords or access management found to be the reason of 30% successful ransomware attacks. Along with that, poor user practices, and malicious websites played their role to become the major attack vectors used for ransomware attacks.

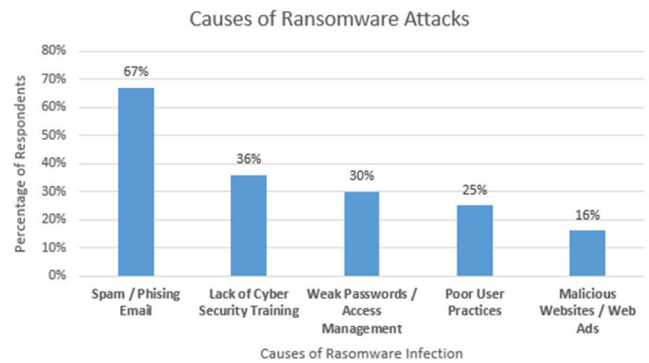


Fig. 1. Causes of Ransomware Attacks (Statista, 2021)

B. Evolution of Ransomware

The first ransomware called PC Cyborg or AIDS Trojan was created by J.L. Popp in 1989 [1]. Since 1989, hundreds of ransomware variants have been created by cyber criminals. In March 2006, Trojan.Cryzip appeared which was used to copy data files in password-protected archive files and delete the original files [3]. In 2007, the locker ransomware first appeared in Russia. Soon, the virus spread to the organizations of US and Europe. In 2008, GPcode.AK was unleashed on the public using a 1024-bit RSA key.

In 2012, a toolkit Citadel was introduced on dark web. Citadel made it easy to distribute ransomware on the machines. In 2014, the CryptoDefense and its improved version CryptoWall were released. CryptoWall used a java vulnerability to deliver malicious advertisements and earned more than 1 million dollars in the single year, 2014. The first JavaScript based Ransomware-as-a-Service was introduced in 2016. Along with that, Petya and WannaCry were the nightmare for organizations in the same year. In 2018, a white hat hacker developed RansomCloud that encrypts cloud

emails in real time. In 2020, Nemty ransomware creators began extorting victims by threatening to publish their private data on public blogs.

C. Detection and Prevention Techniques

Castiglione and Pavlovic, 2019 [4] suggested a technical as well as strategic solution to reduce overall worth of ransomed data by using the dynamic distributed secure storage. The method was given to disperse data on different machines in real time to ensure data distribution on new machines in case of some compromised nodes. They also presented a fast and secure encryption method to ensure data confidentiality on local nodes in a distributed environment.

In 2019, Arabo et al. [5] examined the processes' behavior to determine their malicious nature. By conducting an experiment on 41 benign applications, 7 ransomware and 34 malware samples, they were able to differentiate between benign and malicious files. Their proposed solution required dataset of known ransomware rather than the signature database.

Chew and Kumar, 2019 [6] investigated the traits of significant ransomware malwares: Petya, Cerber, WannaCry, and TeslaCrypt. The authors used the file entropy, the file change, and the manipulation of canary files as indicators of compromise for malware detection. The preventive techniques were implemented against mentioned malwares and their performance was discussed in the paper.

III. RESEARCH METHODOLOGY

In this section, we present the flowchart of methodology used in the research work. Along with that, we also discuss the scope of the work presented.

A. Research Methodology Flowchart

In the first phase, we read ISTR to identify useful trends like top targeted countries by ransomware attacks. We collect data in the form of excel sheets to formulate a small database. In the second phase, we analyze data to identify and resolve conflicts. We process data to formulate information based on the data obtained from multiple authentic reports.

In the third phase, we present the formulated information. Along with that, the comparison of trends from different authentic sources is also presented. In the last phase, the conclusion is given with the discussion on possible research areas that can be explored as future direction.

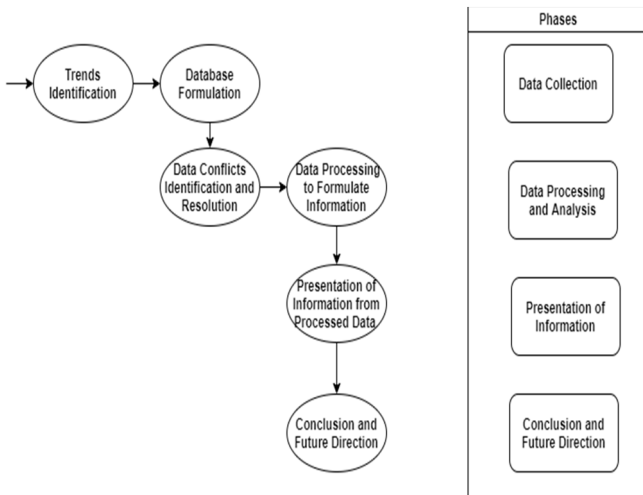


Fig. 2. Flowchart of Research Methodology

B. Scope of the Work

Based on the platform, the ransomware is categorized into five main categories [2]. These categories are PC ransomware, mobile ransomware, cloud based ransomware, Internet of Things (IoT) ransomware, and general ransomware. The general ransomware covers all other platforms except the mentioned ones. We focus on PC ransomware in the presented work. The PC ransomware consider to be the ransomware on employee's official laptops, corporate computers and servers.

We provide survey based on the global data taken from the ISTR for the years 2016-2020. The data from research papers and white papers was not taken into considerations. Also, it is important to note that the data was collected from only McAfee, Kaspersky, and Symantec ISTR. However, several other antivirus companies (like Norton, AVG etc.) also publish their reports. But, data cannot be collected from all the companies due to resource constraints.

IV. MALWARE ATTACK TRENDS

In this section, we presented the malware attack trends taken from threat reports for the last five years. It is important to note that these trends are based on all malware attacks and are not limited to ransomware attacks only.

A. Trends from McAfee ISTR

The McAfee annual reports provided the data on publicly disclosed attacks among different regions, top targeted industries and top attack vectors [8] - [12]. The McAfee data was in the form of graphs, so we mainly extracted the estimated values rather than the exact values.

In publicly disclosed attacks, there were total 4,118 attacks among 5 regions. The region of North America and South America disclosed the most attacks 2,750, while Africa disclosed only 35 attacks, least in all mentioned regions. Asia and Europe disclosed 596 and 659 attacks respectively. Oceania disclosed only 78 attacks.

In top targeted industries, public sector with 1,035 attacks was the favorite target for cyber criminals. Healthcare was attacked 906 times while the education sector had 531 attacks. Finance sector consisted of banking and insurance had 498 attacks. IT consisted of technology, online services and software development had 356 attacks. Entertainment sector had 255 attacks. Retail and Wholesale sector was attacked 236 times by cyber criminals.

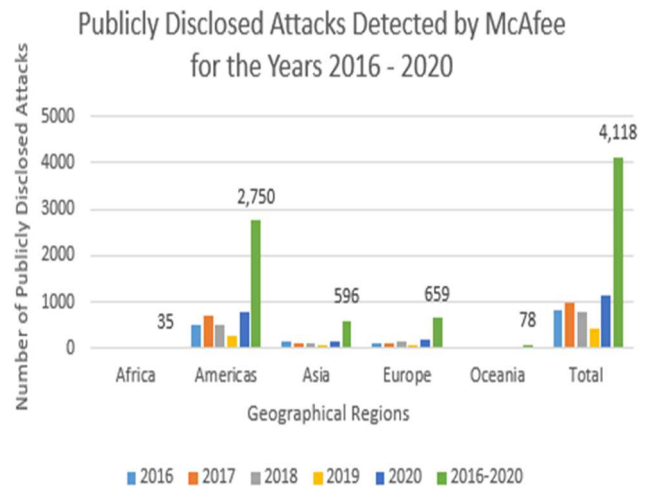


Fig. 3. Publicly Disclosed Attacks for the Years 2016 - 2020

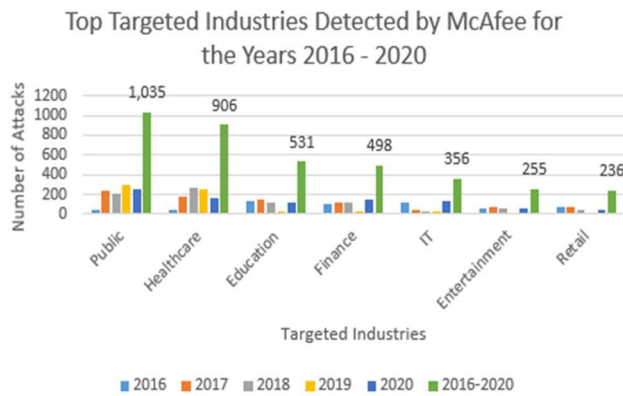


Fig. 4. Top Targeted Industries for the Years 2016 - 2020

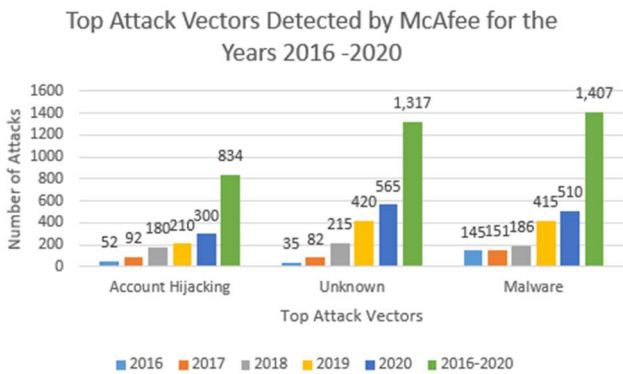


Fig. 5. Top Attack Vectors for the Years 2016 - 2020

In top attack vectors, the malware attack vector was on the highest position with 1,407 attacks. There were 1,317 attacks of unknown attack vector while account hijacking was being done 834 times by cyber criminals. The dangerous aspect in this trend is that the McAfee categorized 1,317 attacks as unknown. This dangerous aspect enforced the corporate sectors to take the most preventive measures against the unknown/zero-day attacks.

B. Trends from Kaspersky ISTR

The Kaspersky quarterly reports provided data on banking targeted countries, top banking malware families, online targeted attacks, and the vulnerable applications percentage [13] – [27]. Kaspersky is a Russian antivirus company with most of its coverage in Asia, and Africa. Due to this fact, we were unable to find much details on American and European countries from Kaspersky ISTR.

In banking targeted countries, Russia was on top with 93,387 attacks detected by Kaspersky. Brazil was the second most targeted country with 81,797 banking sector attacks. Turkey turned out to be the third most vulnerable country with 77,344 attacks. Uzbekistan and Venezuela were among the most targeted countries with 68,690 and 68,246 attacks respectively.

In top banking malware families, Zbot was the nightmare for banking sector with more than 1.5 million attacks. Malware of Nymaim family hit the banking sector more than 0.5 million times. “RTM” was the third most dangerous malware family with 281,877 attacks. Emotet found to be the fourth most dangerous malware with 227,745 attacks. Gozi and SpyEye were among the destructive malwares with 208,177 and 190,476 attacks respectively.

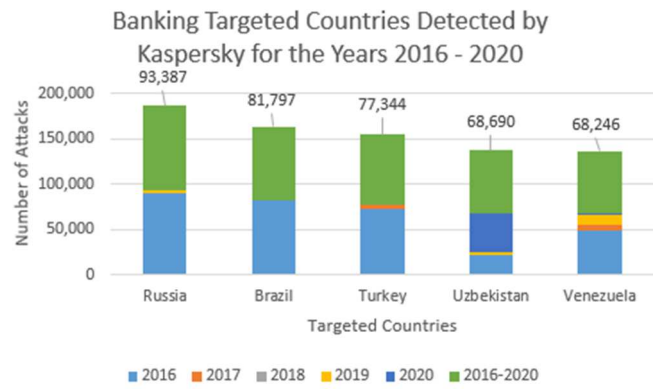


Fig. 6. Banking Targeted Countries for the Years 2016 - 2020

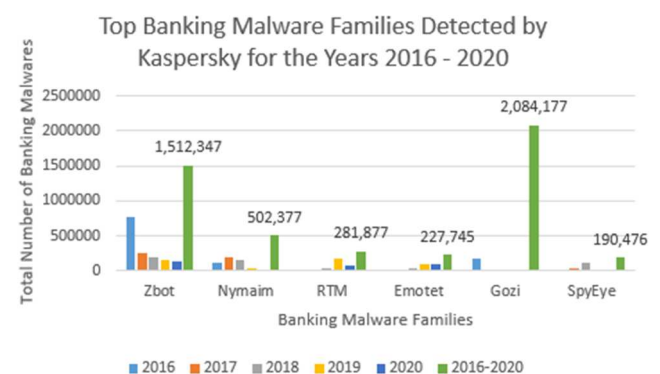


Fig. 7. Banking Malware Families for the Years 2016 - 2020

In online targeted attacks, USA was on top with more than 4.4 billion online attacks detected by Kaspersky. Netherland was the second most targeted country with more than 1.8 billion online attacks. Germany was the third most vulnerable country with 1.0 billion attacks. France was targeted by online attacks with over 359 million times while Russia was targeted via online attacks more than 334 million times by cyber criminals in the last five years.

In vulnerable applications percentage, six application types constituted 99 percent of all vulnerable applications for the years 2016-2020. Office applications were the most vulnerable with the percentage value of 48.56. PDF applications were the least vulnerable with 0.81 percentage value among the mentioned applications. Browsers and Android applications were 25.48 and 16.17 percent vulnerable. Java based applications and Adobe Flash had the vulnerability percentage values of 4.86 and 4.02 respectively.

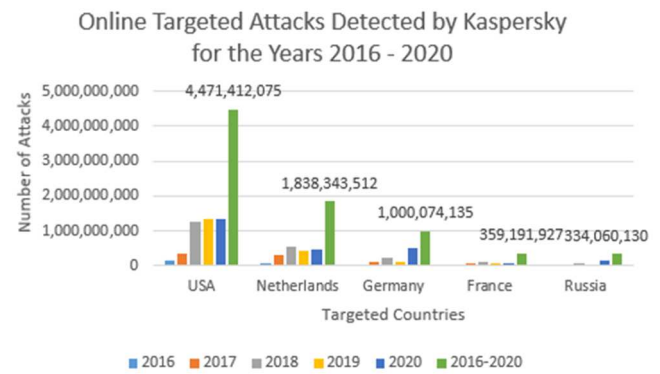


Fig. 8. Online Targeted Attacks for the Years 2016 - 2020

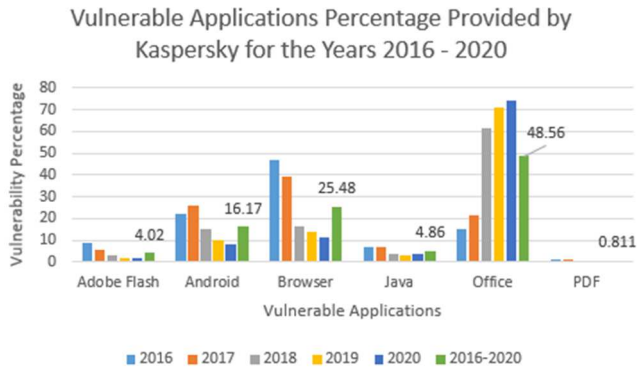


Fig. 9. Vulnerable Applications Percentage for the Years 2016 - 2020

C. Trends from Symantec ISTR

From Symantec reports, we presented the total malwares detected by operating system, and Mac malwares percentage [28] - [30]. Since the data for the years 2019 and 2020 was not available in Symantec ISTR, we were bound to present data for only three years i.e. 2016 - 2018. In order to ensure data integration and data conflicts resolution among the reports, we use average of annual data as an effective tool to compare the Symantec data with the McAfee and the Kaspersky data.

In total malwares detected by operating system, the malwares detected on Windows were much higher than the malwares detected on Mac OS. This was because Windows enjoyed the larger share of PC and more vulnerable than Mac OS. The malwares on Windows were 161 million in 2016, 165 million in 2017, 144 million in 2018, and total 471 million. On the other hand, the malwares on Mac OS were 2.4 million in 2016, 4.01 million in 2017, 4.2 million in 2018, and total 10.6 million.

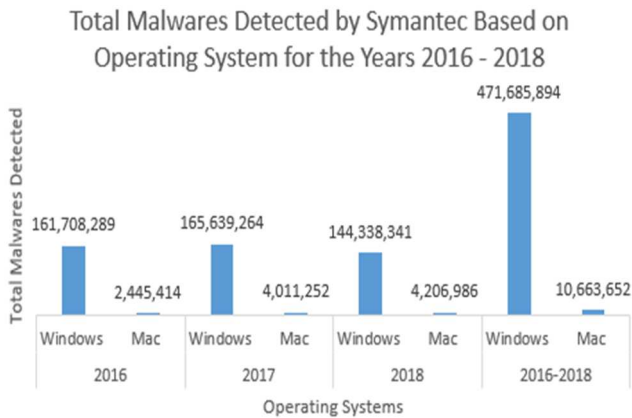


Fig. 10. Total Malwares by OS for the Years 2016 - 2018

In Mac malwares percentage, top five malware presented with their percentage values. The malware OSX.Malcol was the most threatening for users with 9.59 percentage value. The malware W97M.Downloader was the second most dangerous malware with 8.67 percentage value. The malware Heur.AdvML.B found to be the third most threatening to Mac OS users with total 7.5 percentage value. The malware OSX.Malcol.2 positioned at the fourth place with total 6.84 percentage value. Finally, the malware JS.Webcoinminer played its role with total 5.6 percentage value to aid the cyber criminals against the Mac OS users.

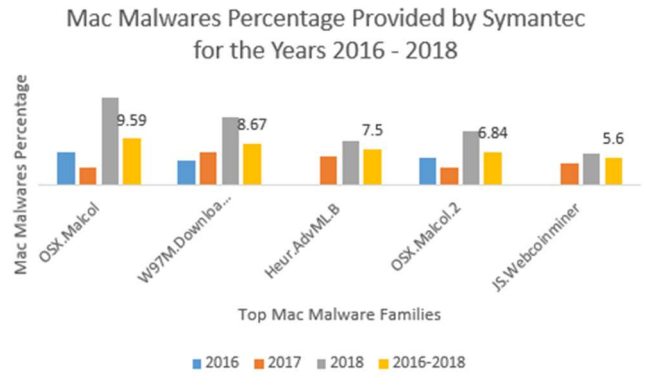


Fig. 11. Mac Malwares Percentage for the Years 2016 - 2018

V. RANSOMWARE ATTACK TRENDS

In this section, we present the top ransomware targeted countries. We provide a comparison of new ransomware families detected by Kaspersky and Symantec. We present the most threatening ransomware families in the last five years. We provide the tables on new malware, total malware, new ransomware, and total ransomware attacks. We also present the estimated probability of occurrence of new and total ransomware attacks.

A. Ransomware Targeted Countries

In ransomware targeted countries, Bangladesh was on top with 92,947 attacks detected by Kaspersky. Uzbekistan was the second most targeted country with 56,156 ransomware attacks. Japan turned out to be the third most vulnerable country with 56,062 attacks. Taiwan, and Croatia were among the top five targeted countries with 47,743 and 38,708 attacks respectively.

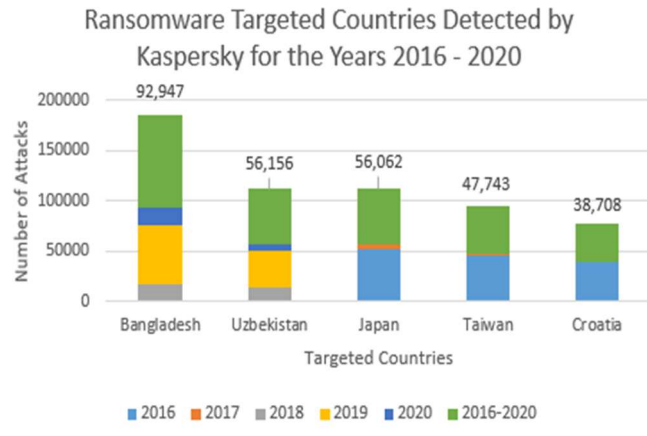


Fig. 12. Ransomware Targeted Countries for the Years 2016 - 2020

B. Ransomware Families

As reported by Clouston et al. [7], there were 505 known ransomware families till the mid-2017. As per Kaspersky reports, total 122 new ransomware families were detected. While Symantec reported 158 new ransomware families. Kaspersky and Symantec only provide the number of new ransomware families in their ISTR. They did not provide the exact names of ransomware families. The year 2016 was the most dangerous in terms of zero-day ransomware attacks. The new ransomware variants were quite low in the year 2018. However, a decent number of ransomware variants were detected in the year 2020.

In top ransomware families, WannaCry was seemed to be the nightmare for the businesses with more than 0.55 million attacks. The CTB-Locker targets the organizations more than 0.36 million times. TeslaCrypt was the third most dangerous ransomware family with 0.25 million attacks. Phny found to be the fourth most dangerous ransomware family with 0.22 million attacks. Locky was the last among the top five ransomware families with 0.16 million attacks on businesses.

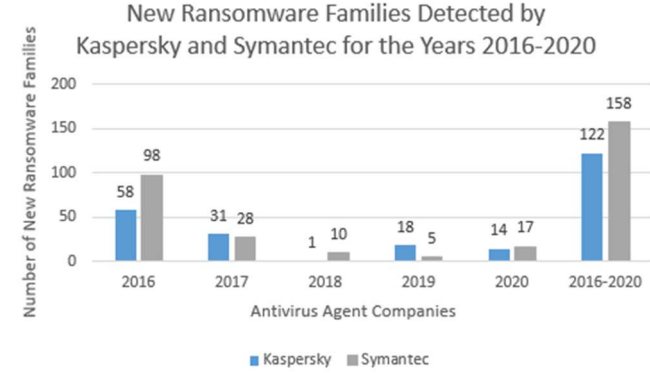


Fig. 13. New Ransomware Families for the Years 2016 – 2020

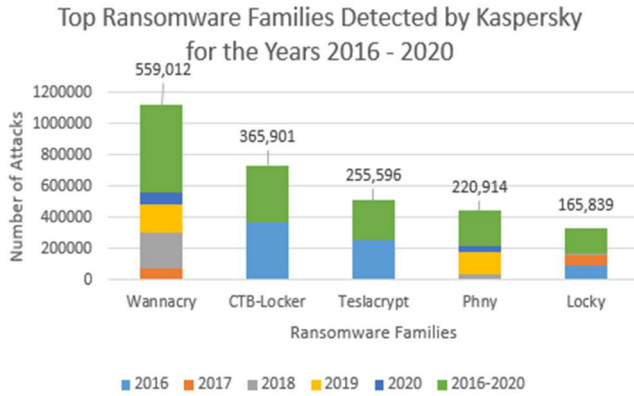


Fig. 14. Top Ransomware Families for the Years 2016 - 2020

C. Data Comparison from McAfee, Kaspersky, and Symantec ISTR

With a comparison on collected data from the reports, we present the new malware attacks detected by McAfee and Symantec, total malware attacks detected by McAfee and Kaspersky, new ransomware attacks detected by Kaspersky and Symantec, and total ransomware attacks detected by McAfee and Symantec.

McAfee detected 1,016 million new malware attacks in the last five years. Symantec detected 2,021 million new malware attacks during the same time period. The McAfee data was based on the estimated values rather than the exact values. Symantec detected new malware attacks were almost doubled the attacks detected by McAfee. The new malware attacks detected by both McAfee and Symantec were 3,037 million, which is a pretty alarming number in terms of attacks.

McAfee detected 17,610 million total malware attacks. While Kaspersky detected 16,988 million total malware attacks. The total malware attacks detected by both McAfee and Kaspersky were 34,598 million. Kaspersky detected 0.27 million new ransomware attacks. Symantec detected 1.30 million new ransomware attacks. The new ransomware

attacks detected by both Kaspersky and Symantec were 1.57 million. The new ransomware attacks detected by Symantec were much higher than the attacks detected by Kaspersky.

McAfee detected 194.04 million total ransomware attacks. Symantec detected 2.87 million total ransomware attacks. The total ransomware attacks detected by Symantec were much lower than the attacks detected by McAfee. Total ransomware attacks detected by both McAfee and Symantec were 196.91 million.

TABLE I. NEW MALWARE ATTACKS 2016-2020

New Malware Attacks	New Malware Attacks (in Millions) for the Years 2016-2020					
	2016	2017	2018	2019	2020	2016-2020
McAfee	126	205	198	281	206	1016
Symantec	357	670	246	424	324	2021
Total	483	875	444	705	530	3037

TABLE II. TOTAL MALWARE ATTACKS 2016-2020

Total Malware Attacks	Total Malware Attacks (in Millions) for the Years 2016-2020					
	2016	2017	2018	2019	2020	2016-2020
McAfee	2400	2870	3220	4120	5000	17610
Kaspersky	1483	2212	4434	4358	4501	16988
Total	3883	5082	7654	8478	9501	34598

TABLE III. NEW RANSOMWARE ATTACKS 2016-2020

New Ransomware Attacks	New Ransomware Attacks (in Millions) for the Years 2016-2020					
	2016	2017	2018	2019	2020	2016-2020
Kaspersky	0.07	0.10	0.03	0.05	0.02	0.27
Symantec	0.24	0.35	0.19	0.26	0.26	1.30
Total	0.31	0.45	0.22	0.31	0.28	1.57

TABLE IV. TOTAL RANSOMWARE ATTACKS 2016-2020

Total Ransomware Attacks	Total Ransomware Attacks (in Millions) for the Years 2016-2020					
	2016	2017	2018	2019	2020	2016-2020
McAfee	30.60	32.70	31.65	31.65	67.44	194.04
Symantec	0.48	0.68	0.54	0.57	0.60	2.87
Total	31.08	33.38	32.19	32.22	68.04	196.91

The presented values are just a depiction of the intensity of cyber-attacks. Using the above mentioned values, we formulate the estimated year-wise occurrence of ransomware attacks to the organizations.

D. Probability of Occurrence of Ransomware Attacks

We formulate the probability of occurrence of new ransomware, and total ransomware attacks. We divide the new ransomware with new malware and multiplied the obtained value with 100 to obtain the probability of occurrence of new ransomware. Similarly, we divide the total ransomware with total malware and multiplied the obtained value with 100 to get the probability of occurrence of total ransomware. The occurrence of total ransomware was much higher than that of new ransomware attacks.

TABLE V. PROBABILITY OF OCCURRENCE OF RANSOMWARE

Probability of Occurrence	Probability of Occurrence of Ransomware Attacks for the Years 2016-2020					
	2016	2017	2018	2019	2020	2016-2020
New Ransomware Occurrence	0.06	0.05	0.05	0.04	0.05	0.05
Total Ransomware Occurrence	0.8	0.65	0.42	0.38	0.71	0.57

From the above table, it is quite clear that every fifth new malware was actually a new ransomware variant in the last five years. On the other hand, about 57 percent malware detected to be as ransomware in the recent years.

VI. CONCLUSION AND FUTURE DIRECTIONS

The antivirus companies publish threat reports on certain intervals. The information provided in these reports is based on their internal data gathered from specific antivirus agent. We have combined the information from multiple threat reports and presented a brief survey on malware attack trends and ransomware attack trends for the last five years.

In malware attack trends, we present the publicly disclosed attacks among 5 geographical regions. We present the top targeted industries and top attack vectors. As per top targeted industries, the public sector found to be the most vulnerable. In banking targeted countries, Russia found to be the most vulnerable. Zbot was the most threatening banking malware family in the recent years. We also present the online targeted attacks and vulnerable applications percentage. We provide a comparison of total malware detected by Windows and Mac OS. We also present the Mac malware percentage.

In ransomware attack trends, we present the ransomware targeted countries where Bangladesh found to be the most vulnerable country. WannaCry was the nightmare for businesses in the recent years. A comparison of new ransomware families detected by Kaspersky and Symantec is also presented. We present a comparison of new malware, total malware, new ransomware and total ransomware reported in the threat reports. Finally, we present an estimated probability of occurrence of new ransomware and total ransomware attacks.

For future work, we would like to use the chunks of presented information to formulate a cyber risk model of ransomware attacks. We would like to develop a model to estimate the potential losses due to ransomware attack to an organization in an industry based on the organization size and the nature of industry.

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